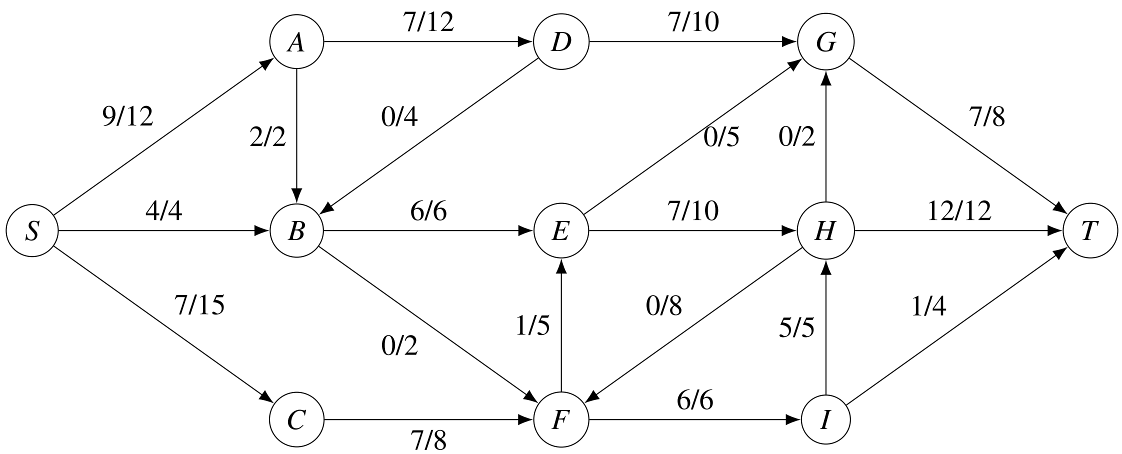
CS 381 Introduction to the Analysis of Algorithms
Spring 2025 Alexandros Psomas Monte Carlo Homework 5

Due Fri Apr 11 at 4:59PM

1. (30 points) **Flow-Up**

   A max-flow algorithm has been performed part-way to achieve the flow in the network shown below. The numbers $f(e)/c(e)$ above each directed edge e represent the current flow and the capacity of that edge respectively. To answer the following questions, you may include photographs of hand-drawn graphs so long as the hand-drawn figures are clear and legible.



   (a) Draw the residual graph for this flow network.

   (b) Run one iteration of Ford Fulkerson. Show the augmenting path and draw the flow network with the updated flow.

   (c) What is the partition of vertices for the minimum cut? Explain how you arrived at your answer.

   (d) What is the max-flow of this network? Give a short (but precise) explanation of why the flow is maximum.

2. (30 points) **Where to Locate?!**

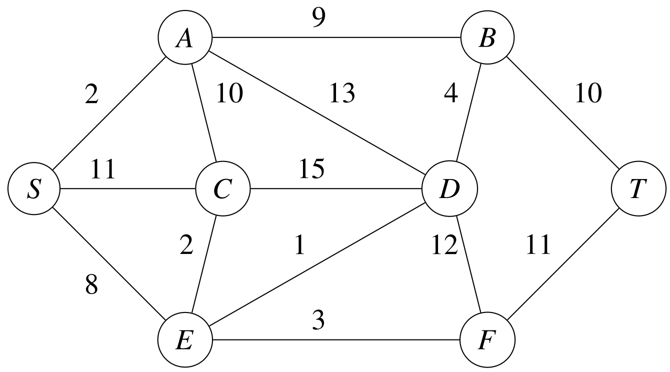
   The Indiana State Government is planning to construct a concrete factory in one of the cities within Indiana. To support the construction, two equipment provider companies will supply heavy machinery: Company S (located in West Lafayette) provides bulldozers and Company T (located in Bloomington) provides cranes. Each bulldozer from Company S must be paired with a crane from Company T in order to be useful for construction. However, due to the massive size of these vehicles, they must avoid roads with tunnels or bridges that have height restrictions lower than their own height.

   Suppose that the Indiana road network between West Lafayette and Bloomington is modeled as an undirected graph $G = (V, E)$, where V represents cities and E represents roads between cities, each with a height limit H_{uv}, indicating the maximum vehicle height that can pass through the road between cities u and v. Let

$h_{s,1}, h_{s,2}, \ldots, h_{s,n}$ be the heights of bulldozers located in West Lafayette and $h_{t,1}, h_{t,2}, \ldots, h_{t,n}$ be the heights of crane located in Bloomington. A bulldozer or crane can only travel along a road (u, v) if its height does not exceed H_{uv}.

The objective is to select a city where the maximum number of valid bulldozer-crane pairs can meet to establish the concrete factory.

For example, consider the following graph. West Lafayette is denoted as S and Bloomington is denoted as T. Let the heights of bulldozers in West Lafayette be $h_{s,1} = 6, h_{s,2} = 5, h_{s,3} = 11$ and the heights of cranes in Bloomington be $h_{t,1} = 3, h_{t,2} = 8, h_{t,3} = 7$. Then the best location to meet is F because all 3 bulldozers and cranes can pair here.



(a) Design an efficient algorithm that takes as input a connected road network and outputs the best location to meet at i.e., one that maximizes the number of bulldozers and cranes that can attend. For simplicity of notation, you may assume West Lafayette is denoted as S and Bloomington is denoted as T in the graph.

(b) Analyze the runtime for your algorithm. For the graph, you may assume that the number of edges is e and number of vertices is m.

(c) Give a proof of correctness.

3. (30 points) **Fencing?**

Old MacDonald has a farm which is represented as an $n \times m$ rectangular grid. Each cell of this grid is either empty or contains a tree.

Being the responsible farmer he is, Old MacDonald wants to build some fences for his trees so that they are protected from the animals. To do this, he needs to place fences to separate tree cells from empty ones. However, he also has the option to plant new trees in empty cells or remove existing trees if it helps reduce the overall cost.

More formally, he can perform the following actions:

- Place a fence between two adjacent cells (one containing a tree and one empty), which costs x dollars.
- Remove a tree from a tree cell (turning it into an empty cell), which costs y dollars.
- Plant a tree in an empty cell (turning it into a tree cell), which costs z dollars.

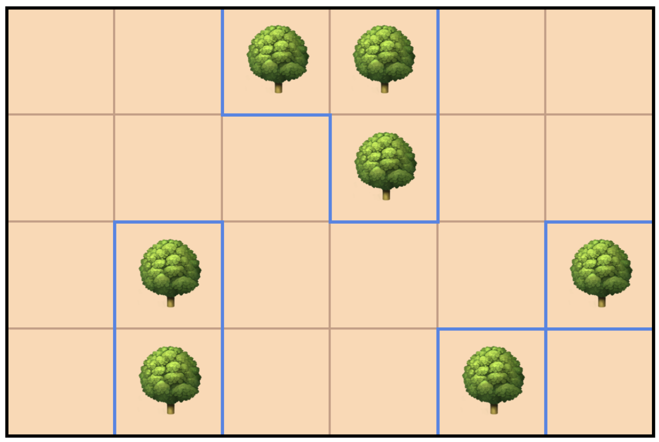
Old MacDonald wants to find the cheapest way to enclose the trees using fences. The number of trees may increase or decrease in the final arrangement, but we don't care about how the changes happen—only about the final layout.

Please note that fences can only be placed between adjacent cells (horizontally or vertically) where one cell has a tree and the other does not. This means that for each cell (i, j), a fence can be placed between (i, j) and the following adjacent cells:

$$\{(i-1, j), (i+1, j), (i, j-1), (i, j+1)\}$$

Also, there is no need to place fences outside the boundaries of the farm, as the edges are already protected. This means that no fence is required between the border cells and the outside of the farm. Lastly, since many animals live on his farm, Old MacDonald ensures that no part of a tree cell is left unfenced. In the final layout, every tree cell must be separated by a fence from all neighboring empty cells.

For example, the following is a valid final layout where the fences are represented by blue borders. A total of 17 fences are used in this layout.



Help Old MacDonald determine an optimal arrangement of trees and fences that minimizes the total cost. Your algorithm should return the minimum cost needed to build fences around the trees in your optimal layout. **Explain your approach and prove the correctness of your algorithm. Write the asymptotic analysis of your runtime (based on n and m).**

4. (10 points) **Party Tables**

   (a) **(1 point)** Create a new chat in ChatGPT-4o and ask it to provide a (succinct) solution to the following problem. Provide a link to your chat (using the "Share Chat" functionality) and a (legible) screenshot of the response. You can include this in Latex using the figure environment.

   If you are working in a study group, everyone in the group can submit the same screenshot (but, of course, write up Part (b) individually).

   "The annual Computer Science Awards Banquet is just around the corner and you have been assigned the delicate task of seat allocation! Fancy dinners always come with weird traditions and this is no exception. For this ceremony, you have access to a sufficiently large number of circular tables (as many as you desire!). n candidates have RSVPed to the event and you are asked to arrange the candidates in these tables any way you want as long as you obey the following rules:

   i. Two people can sit next to each other if and only if the sum of their ages is a prime number!

   ii. All tables must have at least 3 people sitting at them.

Assuming you know everyone's age, and using what you have learned about maximum flow algorithms in class, present a polynomial time algorithm that calculates an appropriate seating arrangement if possible, or state that it is not possible. **Explain your approach and prove the correctness of your algorithm using rigorous arguments. Write the asymptotic analysis of your runtime (based on *n*)**.

Oh, and remember, this is an adult event, so no babies are allowed! In other words, all the guests are between 2 and 116 years old."

(b) **(9 points)** Grade ChatGPT's solution from Part (a). Specifically, if both the algorithm and its proof of correctness work, state that this is the case. If there is a bug in the proof, point to the bug, and argue whether this is a minor bug (e.g., a calculation error) but the proof is more or less correct, or whether it is a major bug (i.e., a crucial step is wrong or missing).